

Jashore University of Science and Technology

Jashore-7408, Bangladesh



Energy-Efficient Route Planning for Low-Cost Battery Powered Vehicles in Urban Areas

Submitted By:

Md. Mushfiqur Rahman

200121

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Supervisor:

Monishanker Halder

Assistant Professor

Dept. of Computer Science and Engineering

Jashore University of Science and Technology

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[Thesis approval text]

Signature of the Student

Md. Mushfiqur Rahman

Signature of the Supervisor

Signature of the Co-Supervisor

Dedication

This thesis is dedicated to my family and mentors, whose unwavering support and encouragement have been my guiding light. Their belief in my abilities and constant inspiration made this work possible.

Acknowledgments

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Abstract

With the rapid growth of population, transportation demand has significantly increased in the urban areas of developing countries like Bangladesh. Major cities are facing transport related challenges like affordable public transport. Public transports are generally shared by the general public. To meet this demand both cost-efficient and environment friendly solutions are needed and in this scenario battery powered vehicles like three-wheeler easy bikes, electric rickshaws and electric vehicles became a part of the urban transportation system as an alternative to the traditional engine vehicles. In recent times the number of battery-powered vehicles has massively increased for its accessibility and comfort. But these Battery Powered vehicles are limited by a short operational range (60–100 km per charge), leading to frequent downtime. This thesis proposes an energy-efficient route planning solution to solve this by minimizing consumption and extending coverage. The approach involves developing a high-fidelity traffic simulation model of Jashore using SUMO, incorporating real-world vehicle data, and validating a Vehicle-to-Infrastructure (V2I) data collection architecture that captures dynamic energy telemetry via Roadside Units (RSUs). This dataset will be used to train a machine learning (ML) algorithm to predict energy costs across road segments, which will then optimize routes based on energy conservation. This software-based strategy is expected to significantly increase vehicle uptime and daily operational range, substantially benefiting the sustainability and efficiency of urban transport in Bangladesh.

Chapter 1

Introduction

Chapter 1: Introduction

In this paper we are working in the perspective of the city Jashore of Bangladesh where these battery powered vehicles are the most dominant share of the public transportation system. Although these battery powered three wheelers became the most popular transport it has a big drawback of its uptime. As these vehicles are powered by batteries they give a coverage of 60-100 km per charge. So we tried to work on the energy-efficient route planning for these battery-powered vehicles to more optimize their usage and extend their coverage. We have tried to find the most energy-efficient route to reach a destination. For this work we modeled these battery-powered vehicles within a simulation environment using SUMO (Simulation of Urban Mobility). We have made a dataset by collecting their real-world data on vehicle configurations, charging patterns, usage times and daily costs from actual owners and drivers. This dataset has important parameters like battery charging times, uptimes, energy consumption and energy demands. Then we have used machine learning algorithms to optimize the routes for these vehicles. The model will analyze energy consumption and battery status data and generate optimized routes that minimize the energy usage and increase the uptime for the vehicles. This work will be a great tool for extending the range of battery powered vehicles. It will also reduce the cost and give more uptime. So it will greatly help the battery-powered vehicle sectors in the urban areas of Bangladesh.

1.1 Background

Urban population growth in developing countries has led to a rapid rise in transportation demand, especially in city centers. Many cities struggle to provide affordable and sustainable public transportation for their expanding populations. In response to these challenges, environmentally friendly and cost-efficient alternatives to traditional fossil-fuel vehicles are being sought worldwide. One promising development is the adoption of electric, battery-powered vehicles for public transport, which produce no tailpipe emissions and can offer a cheaper mode of travel for the masses.

In Bangladesh and similar contexts, low-cost battery-powered three-wheeler vehicles (such

as easy bikes and electric rickshaws) have become an integral part of urban transportation. These vehicles are popular due to their accessibility, low operating cost, and ease of use, effectively filling gaps in the public transit system. However, a significant drawback of these electric vehicles is their limited driving range or uptime on a single battery charge – typically around 60–100 km before requiring lengthy recharging. This constraint leads to frequent downtime for charging, reducing the vehicles’ availability and earning potential for drivers. The limited range, combined with a lack of strategic route planning, means that energy is often used inefficiently during daily operations. There is a clear need for optimizing the routes of these battery-powered vehicles to minimize energy consumption on trips. By finding energy-efficient routes for a given destination, the distance and energy wasted can be reduced, effectively extending the vehicles’ usable range per charge and improving their overall service uptime.

1.1.1 Definition

1.1.1.1 Battery-Powered Three-Wheeler

A small motorized vehicle with three wheels that runs on an electric battery instead of a combustion engine. This category includes vehicles like easy bikes and electric rickshaws, which are used as low-cost public transport in many Asian cities. These vehicles produce no direct emissions and typically carry several passengers at a time.

1.1.1.2 Easy Bike

A locally used term in Bangladesh for a type of electric three-wheeler taxi. Easy bikes are battery-powered vehicles, usually seating around 4–6 passengers, and are widely used for short-distance transportation in urban areas. They have a relatively low top speed and are valued for being affordable to operate and environmentally friendly.

1.1.1.3 Electric Rickshaw

An electric three-wheeled rickshaw, similar in purpose to an easy bike but often resembling the traditional pedal-powered rickshaw in design. It is powered by a rechargeable battery and can carry a driver and a few passengers. Electric rickshaws provide a motorized alternative to cycle rickshaws, reducing human effort and increasing speed while still being emission-free at the point of use.

1.1.1.4 Uptime

In the context of battery vehicles, uptime refers to the amount of time or distance that a vehicle can operate before the battery is depleted. Higher uptime means the vehicle can serve longer routes or more trips between charges. Uptime is directly limited by battery capacity and energy consumption rate; for example, an uptime of 80 km means the vehicle can travel approximately 80 km on a full charge.

1.1.1.5 Energy-Efficient Route Planning

The process of selecting a travel route that minimizes energy consumption for a vehicle, rather than simply the shortest or fastest route. For electric vehicles, an energy-efficient route might avoid factors that increase energy use, such as steep hills, heavy traffic (which causes stop-and-go inefficiency), or high-speed roads that drain the battery faster. The goal is to reach the destination using the least amount of battery power, thus extending the vehicle's effective range.

1.1.1.6 Charging Station

A dedicated facility or setup where electric vehicles can recharge their batteries. Charging stations for easy bikes and electric rickshaws are often located at garages or key points in the city. They supply electric power (typically from the grid) to restore the vehicle's battery charge. The availability and placement of charging stations directly affect how conveniently vehicles can be recharged during or between service periods.

1.1.1.7 Simulation of Urban Mobility (SUMO)

An open-source traffic simulation software package that allows researchers to model and simulate road networks, vehicle movement, and traffic behavior. SUMO provides a virtual environment to test traffic scenarios, route planning strategies, and vehicle performance without real-world trials. In this project, SUMO is used to create a model of Jashore's urban road network and to simulate the behavior of battery-powered vehicles under different routing strategies.

1.1.1.8 Traffic Control Interface (TraCI)

A software interface that enables external programs to interact with a SUMO simulation in real time. TraCI (often used with Python) allows for retrieving simulation data (such as a vehicle's speed, position, or battery charge at each timestep) and for dynamically controlling aspects of the simulation (such as vehicle routing or traffic signals). It was used in this study to extract vehicle telemetry data during simulations for analysis and to implement dynamic interactions with the simulated vehicles.

1.1.1.9 Machine Learning

A branch of artificial intelligence involving algorithms that learn patterns from data and make predictions or decisions. In the context of this thesis, machine learning techniques are employed to analyze complex relationships between route characteristics and energy consumption. By learning from simulation results (and potentially real-world data), the model can predict which routes are likely to be most energy-efficient for the vehicles.

1.1.1.10 Roadside Unit (RSU)

A fixed electronic device placed along the road infrastructure that can communicate wirelessly with vehicles. RSUs are part of a V2I system: they can detect or receive data from passing vehicles (such as location, speed, or battery status) and relay that information to a central server or traffic management system. In this project, simulated RSUs are used to collect real-time data

from electric vehicles in the network, demonstrating how vehicle data could be gathered for route optimization in a real deployment.

1.1.1.11 Vehicle-to-Infrastructure (V2I) Communication

A form of wireless data exchange between vehicles and road infrastructure. V2I enables vehicles to send information (like telemetry or location data) to infrastructure nodes (such as RSUs or smart traffic lights) and receive information in return. This technology is leveraged in advanced transport systems to gather live data on vehicle performance and traffic conditions. In the scope of this study, V2I communication allows the system to obtain up-to-the-minute information on each vehicle's battery level and position, which can be used to inform or adjust route planning for better energy efficiency.

1.2 Problem Statement

The central problem addressed in this research is the limited range and frequent downtime of low-cost battery-powered vehicles in urban areas, caused by inefficient energy use on their routes. Currently, drivers of electric three-wheelers often follow conventional routes without considering battery optimization, leading to rapid depletion of charge and the need for frequent recharging. This lack of energy-aware route planning reduces the vehicles' effective service time and reliability. The challenge, therefore, is to develop a method for planning vehicle routes that minimizes energy consumption, so that these electric vehicles can travel longer distances on a single charge and spend more time in service.

1.3 Objectives

- To develop a realistic simulation model of an urban transport network (using SUMO) that incorporates low-cost battery-powered vehicles and their operational parameters.

- To identify and analyze the factors that affect energy consumption and range for battery-powered three-wheelers on various routes.
- To design an optimization algorithm (using machine learning and/or heuristic methods) that can determine the most energy-efficient route for a vehicle between any given origin and destination.
- To implement a vehicle-to-infrastructure (V2I) data collection framework using Roadside Units (RSUs) and a central server for real-time telemetry monitoring.
- To evaluate the proposed route planning approach through simulation experiments and data analysis.

1.4 Research Questions

- RQ1: What are the key factors that influence the energy consumption and limited range of low-cost battery-powered three-wheeler vehicles in urban environments?
- RQ2: How can route planning be optimized to minimize energy usage and extend the travel range of battery-powered vehicles for a given origin–destination trip in a city like Jashore?
- RQ3: To what extent can the implementation of an energy-efficient routing strategy improve the operational uptime (distance per charge and service hours) of these electric vehicles compared to conventional routing practices?
- RQ4: How can real-time data and vehicle-to-infrastructure (V2I) communication (e.g., via Roadside Units) be leveraged to support and enhance energy-efficient route planning for battery-powered urban vehicles?

1.5 Significance of the Study

The outcomes of this study have practical significance for sustainable urban transportation in developing regions. By extending the range and uptime of battery-powered public transport vehicles, the proposed route optimization approach can directly benefit both drivers and passengers. Drivers will be able to serve more trips per charge, increasing their daily income and reducing downtime spent at charging stations. Passengers, in turn, gain access to more reliable and potentially faster service, since vehicles can remain operational for longer periods without needing to stop for recharge. In the context of Bangladesh's cities (such as Jashore), where electric three-wheelers are a major mode of transport, these improvements can enhance the overall efficiency and affordability of the urban transit system.

Beyond the immediate local benefits, the study contributes to broader goals of energy efficiency and environmental sustainability. More efficient use of battery power means less strain on the electrical grid and longer battery lifespans, which can reduce operational costs and the environmental impact associated with electricity generation. This research also demonstrates how data-driven strategies (combining traffic simulation, real-world data, and machine learning) can address transportation infrastructure challenges. The integration of vehicle-to-infrastructure communication in the proposed solution provides a forward-looking example of smart city technology improving transportation. In academic terms, the work fills a knowledge gap by focusing on route optimization for informal electric vehicles, offering insights that can be applied to similar transportation challenges in other cities and laying groundwork for future research in intelligent transport systems.

Chapter 2

Literature Review

Chapter 2: Literature Review

The field of Electric Vehicle (EV) route planning has evolved rapidly, moving from foundational algorithmic work focused on simple energy minimization to complex, multi-objective systems that integrate real-time data, battery dynamics, and urban infrastructure. This review synthesizes the contributions of recent work, organized chronologically by the complexity of the models and objectives addressed.

Early Algorithmic Foundations

Foundational work by [7] established the challenge of large-scale EV routing by modeling **energy recovery (regenerative braking)** and battery constraints. By adapting Dijkstra's algorithm and integrating contraction hierarchies, they achieved ultra-fast route planning with a query time complexity of $O(n \log n + m)$. Similarly, research on routing systems to extend driving range [23] demonstrated that using **linear energy consumption models** based on speed and slope, combined with Dijkstra's algorithm for energy minimization, results in more accurate range predictions and reduces range anxiety.

Multi-Objective and Data-Driven Advancements

The mid-to-late 2010s saw a pivotal shift towards **multi-objective and data-driven planning**. For instance, MORP [25] proposed a comprehensive approach to simultaneously minimize travel time, energy, charging cost, and 'range anxiety' by integrating sophisticated **ARIMA and kriging models for traffic prediction** and solving the problem using the adaptive epsilon constraint method. This simulation, verified in SUMO, showed marked improvements over baseline strategies.

Concurrent studies focused on integrating routing with charging infrastructure. Work on optimal routing and scheduling of charge for EV fleets [2] introduced a **Differential Evolution Algorithm** to optimize route assignment, time windows, and battery state-of-health (SOH), demonstrating a reduction in operational costs. Other researchers addressed new network features: one study [15] explored **Dynamic Wireless Charging** for EVs, modeling a mobile en-

ergy disseminator to extend range, while another [16] focused on the potential of '**green wave**' **traffic signals** to reduce stops and save energy and time.

Data integration expanded to include financial factors, such as research [18] that incorporated **real-time electricity spot prices** into route and charging planning to minimize total cost. The importance of vehicle dynamics was also emphasized, particularly for low-speed autonomous EVs, where **energy-efficient path planning** [20, 21] using graph-based modeling and rolling resistance-aware energy models demonstrated that shortest paths are often *not* the most energy-saving. Other work developed an innovative EV route-mapping heuristic [26, 27] based on smart grid concepts.

Integrated Systems and Advanced AI

Recent literature demonstrates a trend towards highly integrated, real-time, and AI-driven solutions. Several studies focused on the **Vehicle Routing Problem (VRP) with charging constraints** [4, 17, 8], addressing fleet-level delivery schedules, partial recharge strategies, and optimizing single-recharge routes for efficiency. These works frequently employ advanced metaheuristics like **Large Neighborhood Search and Granular Tabu Search** to manage complexity.

The refinement of energy models to include real-world factors became a major theme. Research incorporated **topography and battery aging** into cost functions [24], showing the value of multi-criteria route planning that balances energy, time, and battery health under varying terrain. Other approaches modeled the **impact of multiple speed states (city vs. highway)** [3] and developed enhanced **CNN-LSTM models for energy consumption prediction** [9], proving that accurate energy forecasting is vital for path planning efficiency. Furthermore, papers [14] incorporated **real-time ambient temperature** into routing, recognizing its effect on battery performance.

The integration with smart-city infrastructure and user experience is prominent in current work. New systems proposed **charging-warning and path-planning** using constraint satisfac-

tion [5, 6] to alert drivers before battery depletion. Furthermore, studies leveraged **Deep Reinforcement Learning (DRL)** [12] to learn optimal routing and charging decisions in dynamic, changing traffic environments, minimizing travel time and waiting. The focus on multi-vehicle coordination also grew, with strategies developed for **collaborative charging scheduling and route planning** [31] aimed at reducing grid stress and queueing at charging stations. Research on **fuzzy classification** [29, 30] and **Ant Colony Optimization (ACO)** [28, 19] shows the continued use of advanced heuristics to handle route planning in large, complex networks.

State-of-the-art research continues to tackle large-scale complexity and emerging infrastructure. This includes **proactive route planning** [22] with optimized partial and fast charging stops, and the introduction of systems for **Battery Electric Buses (BEVs) on electrified roads** [1], where paths are optimized to maximize the use of charging-in-motion segments. Furthermore, research focused on the planning of **unmanned charging vehicles** [13] and the use of **integrated environment-sensing path planners** [10, 11] for electric unmanned sanitation vehicles.

Chapter 3

Methodology

Chapter 3: Methodology

3.1 Research Design

The research follows a sequential, modular design that integrates simulation, real-time data logging, and data analysis. The flowchart 3.1 below illustrates the step-by-step approach used to develop and evaluate energy-efficient routes for battery-powered vehicles in Jashore city.

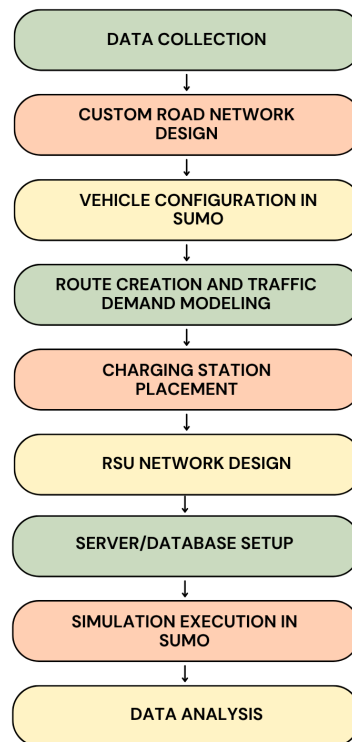


Figure 3.1: Research Design flow

1. Data Collection

The process begins with collecting real-world specifications for battery-powered three-wheelers including Easy Bikes (ER-02B), Small Easy Bikes (V12), and Electric Rickshaws (V8). Data was gathered from manufacturers and verified with local operators, covering vehicle dimensions, battery capacity, charging behavior, and energy consumption profiles.

2. Custom Road Network Design

A custom road network was constructed 3.3 using OpenStreetMap data and refined in NetEdit to capture seven critical junctions in Jashore. The road geometry, junction curvature, and directional flow were tailored for simulation fidelity.

3. Vehicle Configuration in SUMO

Each vehicle model was configured within SUMO using realistic parameters such as acceleration, braking, mass, battery capacity, and efficiency. These configurations ensured accurate energy modeling during route simulation. 3.1 shows the configuration of the vehicles.

4. Route Creation and Traffic Demand Modeling

Five distinct routes were designed to reflect common urban travel patterns, varying in distance and complexity. In 3.2 vehicle trips were synchronized in simulation to enable comparative analysis of energy use across different paths.

5. Charging Station Placement

Ten strategically placed charging stations 3.3 were modeled in the simulation to reflect actual garage and charging locations in Jashore. Each station was configured with specific power ratings, efficiency values, and charging delays.

6. RSU Network Design

Seven Roadside Units (RSUs) were virtually deployed to collect vehicle telemetry such as speed, battery state, and location. Each RSU covered a 500-meter radius and was positioned to ensure full network visibility and redundancy. In 3.6 its shown partially.

7. Server/Database Setup

A FastAPI-based RESTful server backend was developed to ingest RSU data in real-time. It validated and stored telemetry data in a lightweight SQLite3 database across three tables — vehicle logs, RSU status, and raw telemetry.

8. Simulation Execution in SUMO

The entire simulation including vehicle movement, charging behavior, and RSU tracking was executed in SUMO 3.4. Simulation step size was set to 0.5 seconds for high-resolution tracking of battery consumption.

9. Data Analysis

Post-simulation data was exported using custom Python scripts. Energy usage, route efficiency, and charging behavior were analyzed to identify optimal energy-saving routes. These insights serve as the foundation for energy-efficient path planning algorithms.

3.2 Data Collection

In Jashore Three-wheeler battery powered vehicles are the most common public transport. Among them we have selected three categories of battery powered vehicles including easy bikes, small easy bikes, electric rickshaws. For these three categories we have selected three models sequentially ER-02B for easy bikes, V12 for small easy bikes, V8 for electric rickshaws. These models are commonly operated in Jashore city. So we have collected their detailed specifications from both manufacturers and local operators and owners.

3.3 Implementation

3.3.1 Environment Setup

For this work we used **Eclipse SUMO** (Simulation of Urban Mobility) for the simulation environment. For designing the network we have used **NetEdit** and For real-time data extraction **TraCI** (Traffic Control Interface) is used. This simulation environment was set up in a **Windows 11** operating system with environment variable `SUMO_HOME` set to enabled. For exporting real-time data from TraCI to an **Excel file** we have used the **pandas** library. Here all the configuration files are written in **python 3**.

3.3.2 Custom Road Network Creation

We have created a custom road network 3.3 of some important points of the city **Jashore** of Bangladesh. Jashore is a busy city as it is an important transport corridor of Bangladesh. We have used the **OpenStreetMap (OSM)** 3.2 to follow the base road layout. We have designed the road network in **NetEdit** keeping the necessary roads. We have tried to match the road geometry with the real-world curvature of the roads. Our custom road network includes **7 most important junctions** of the Jashore city: Palbari, Dhormotola, Chachra, Doratana, New Market, Monihar and Muroli. These junctions are connected through **22 directional road edges**. For realistic corner radius we have set the parameters `junctionCornerDetail = 5` and `limitTurnSpeed = 5.50 m/s`. As the battery powered vehicles are recharged in garages, to represent them we have placed some **charging stations** in the custom road network with an additional xml file.

3.3.3 Vehicle Configuration in SUMO

We have configured the vehicles with our collected parameters in the sumo configuration file. We have added General Attributes like `vClass`, `acceleration`, `deceleration`, `sigma` (driver imperfection), `length`, `width`, and `maximum speed`. Battery Device Parameters like `Capacity`, `initial`

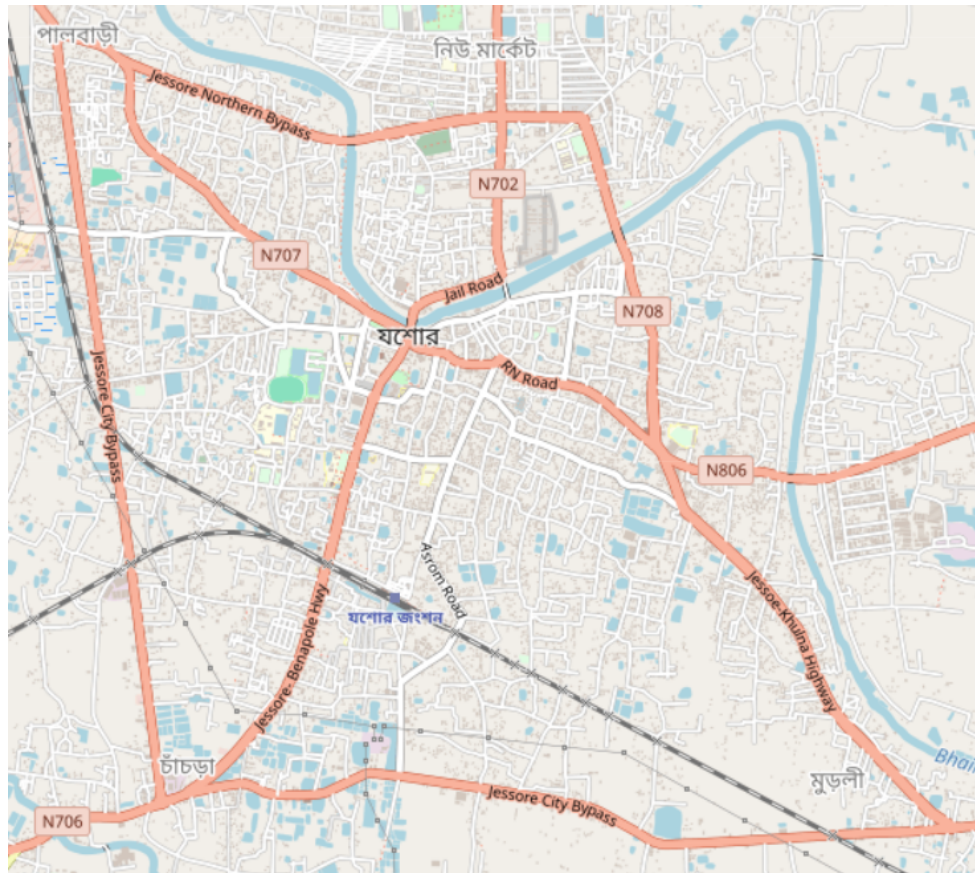


Figure 3.2: Map of Jashore City

charge, vehicle mass, maximum power, propulsion and recuperation efficiency and Operational Attributes like Maximum loading capacity, climbing capacity, braking system type, wheelbase, track width, tire sizes, body material, and electrical specifications. 3.1 shows the parameters that have been used for configuring the vehicles

3.3.4 Route Creation and Traffic Demand Modeling

We have configured a total of **5 unique vehicle routes** in the custom road network. Each of these routes represents a distinct directional flow. We have designed the routes in such a way that it covers major inter-junction travel paths. We have defined the route considering the real-time traffic patterns. Here we mainly covered three types of routes: routes representing **short point-to-point trips**, routes covering **medium-distance travel** with junction transitions and

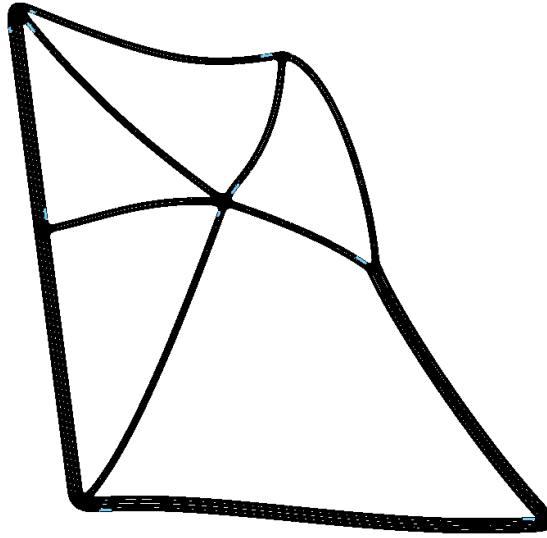


Figure 3.3: Custom Road Network

routes covering **longer trips** across multiple junctions and road segments. We have configured each route pattern to **three different electric vehicle types**: Easybike of model ER-02B, Small Easybike of model V12, and Electric Rickshaw of model V8, resulting in **15 individual vehicle entries** in the route file. In this configuration all of the vehicles depart at simulation time 0.00s which ensures simultaneous operation for comparative analysis. The route file includes departure times, vehicle type parameters and route IDs.

3.3.5 Charging Station Placement

We have placed charging stations representing the local garage locations. In our custom road network of Jashore we have placed a total of **10 charging stations**^{3.3} in a strategic way to the major corridors to support the operation of battery powered vehicles. Here we have placed the stations at key hubs and important junctions including **Palbari, Doratana, New Market, Dhormotola, Chachra, Muroli, and Monihar**, covering both directions where needed. Most stations operate at **2200 W** with **0.9 efficiency**, while two legacy points use **1500 W**. All stations occupy **5–10 m** lane segments, include a **5 s** charge delay, and are restricted to `evehicle` class vehicles.

Table 3.1: Parameters for vehicle configuration

Parameter	Easy Bike (ER – 02B)	Small Easy Bike (V12)	Electric Rickshaw (V8)
vClass	evehicle	evehicle	evehicle
accel (m/s ²)	1.0	1.0	1.0
decel (m/s ²)	2.5	2.5	2.5
sigma (s)	0.5	0.5	0.5
length (m)	2.87	2.37	2.37
width (m)	1.05	0.90	0.90
maxSpeed (m/s)	13.9	7.78	7.78
guiShape	passenger	passenger	passenger
emissionClass	HBEFA3/PC_G_EU4	HBEFA3/PC_G_EU4	HBEFA3/PC_G_EU4
has.battery.device	true	true	true
device.battery.capacity (J)	1008000	672000	672000
device.battery.initialCharge (J)	1008000	672000	672000
device.battery.vehicleMass (kg)	700	500	500
device.battery.maximumPower (W)	1000	800	800
device.battery.propulsionEfficiency	0.85	0.85	0.85
device.battery.recuperationEfficiency	0.80	0.80	0.80
maxLoadingCapacity (kg)	500	300	300
continuedTripMileage (km)	120-150	100-120	100-120
chargeTime (hours)	6-8	6-8	6-8
numPassengers	5 seated	2 adults, 1 kid	2 adults, 1 kid
climbingCapacity (degrees)	30	10	10
brakeMode (F/R)	Drum brakes	Drum brakes	Drum brakes
overallSize (mm)	2870x1050x1780	2370x900x1140	2370x900x1360
carHopper (mm)	1370x1050x870	-	-
wheelbase (mm)	2160	1740	1740
track (mm)	910	790	790
backTireSize	3.25-16	2.75-14	2.75-14
frontTireSize	3.25-16	2.75-14	2.75-14
bodyMaterial	Resin cloth	Cathode electrophoresis	Cathode electrophoresis
inputVoltage (V)	220V, 50HZ	220V, 50HZ	220V, 50HZ
outputVoltage (V)	DC, 60V	DC, 48V	DC, 48V

3.3.6 Simulation Execution in SUMO

The simulation was executed using sumo-gui 3.4 for visual inspection and sumo for performance runs. The configuration file here combines the custom road network, traffic demand (routes), and charging station definitions. **Simulation step length** was set to **0.5 seconds** for battery consumption tracking. We also defined delays to ensure proper initialization. Three electric vehicle types Easybike ER-02B, Small Easybike V12, and Electric Rickshaw V8 were simulated in the custom road network according to predefined route patterns.

Table 3.2: Created routes in SUMO

Route ID	Route Places	Distance (km)
R1	Palbari → Doratana	2.41
R2	Doratana → Dhormotola → Palbari	3.42
R3	Doratana → New Market → Palbari	3.62
R4	Doratana → Muroli → Chachra → Dhormotola → Palbari	11.96
R5	Palbari → Chachra → Doratana	6.97

Table 3.3: Charging Station Placement

Station ID	From → To	Lane ID	StartPos–EndPos (m)	Power (W)	Efficiency	Charge Delay (s)
cs_er02b	Palbari → Dhormotola	E3_0	0–5	2200	0.9	5
cs_palbari_west	Palbari → Doratana	E0_0	0–10	2200	0.9	5
cs_v12	Doratana → Chachra	E4_0	0–5	1500	0.9	5
cs_doratana_central	Doratana → New Market	E1_0	0–10	2200	0.9	5
cs_v8	Palbari → New Market	E2_0	0–5	1500	0.9	5
cs_newmarket_out	New Market → Palbari	-E2_0	0–10	2200	0.9	5
cs_dhormotola_out	Dhormotola → Palbari	-E30_0	0–10	2200	0.9	5
cs_chachra_out	Chachra → Muroli	E5_0	0–10	2200	0.9	5
cs_muroli_out	Muroli → Chachra	-E5_0	0–10	2200	0.9	5
cs_monihar_area	Monihar → Doratana	E7_0	0–10	2200	0.9	5

3.3.7 Real-Time Data Extraction Using TraCI

The **TraCI API** was used to extract vehicle data during simulation. For this we have initialized **python connected to SUMO** using the `sumocfg` configuration file. After Starting the simulation we collected **vehicle speed**, **waiting time**, **battery capacity**, and **actual battery charge** at each simulation step. We have logged the data in an **excel file** after the simulation ends 3.4.

3.3.8 Server Implementation

The system's backbone is a high-performance **RESTful API server** implemented using the **FastAPI** framework in Python. This server acts as the central hub for storing real-time telemetry data received from Roadside Units (RSUs) 3.5.

Key Technology and Architecture

The architecture follows a standard tiered design, separating the **API Layer** (FastAPI/Pydantic validation), **Business Logic** (data processing), and **Data Access Layer** (SQLite operations).



Figure 3.4: Simulation execution in SUMO

Database Schema and Data Flow

The server uses **three primary tables** to store V2I communication data:

- `rsu_vehicle_logs`: Main table for real-time telemetry from vehicles via RSUs, including speed, battery charge/capacity/percentage, and RSU/vehicle identifiers.
- `rsu_status`: Tracks RSU operational metrics like vehicle count and transmitted records.
- `vehicle_logs`: Maintains compatibility with direct TraCI (Simulation of Urban Mobility) data collection.

The POST `/ingest_rsu` endpoint is the core mechanism for data ingestion, receiving **batched telemetry** from RSUs for efficient network and database usage. Batch processing (up to 100 records) was found to yield a **93%** reduction in per-record overhead.

Table 3.4: Data extraction table from TraCI (Partial Example)

Vehicle ID	Speed (m/s)	Battery Charge (J)	Battery Capacity (J)	Time (s)
t_10_Easybike_ER-02B	0	504000	1008000.00	0.5
t_11_Easybike_ER-02B	0	504000	1008000.00	0.5
t_1_Easybike_ER-02B	0	504000	1008000.00	0.5
t_2_Easybike_ER-02B	0	504000	1008000.00	0.5
t_8_Easybike_ER-02B	0	504000	1008000.00	0.5
t_9_Easybike_ER-02B	0	504000	1008000.00	0.5
t_10_Easybike_ER-02B	0.419568296	503999.9958	1008000.00	1
t_11_Easybike_ER-02B	0.382456377	503999.9962	1008000.00	1
t_1_Easybike_ER-02B	0.492534332	503999.9951	1008000.00	1
t_2_Easybike_ER-02B	0.482148213	503999.9952	1008000.00	1
t_8_Easybike_ER-02B	0.491046696	503999.9951	1008000.00	1
t_9_Easybike_ER-02B	0.380344554	503999.9962	1008000.00	1
t_10_Easybike_ER-02B	0.897795405	503999.991	1008000.00	1.5
t_11_Easybike_ER-02B	0.8686672	503999.9913	1008000.00	1.5

3.3.9 RSU Sensor Placement and Data Extraction

Roadside Units (RSUs) are the crucial link in the Vehicle-to-Infrastructure (V2I) system, responsible for detecting and forwarding vehicle telemetry to the central server.

RSU Network Design and Coverage

- **Deployment:** 7 RSUs were strategically positioned in the Jashore city area to ensure comprehensive coverage of key junctions (Palbari, Doratana, Dhormotola, etc.).3.6

Table 3.5: Key Technology and Architecture Components

Component	Technology	Role
Backend	FastAPI (Python 3.x)	High-performance, asynchronous web framework with automatic Pydantic data validation
Web Server	Uvicorn	An ASGI server that efficiently handles concurrent data streams from multiple RSUs
Database	SQLite3	Lightweight, serverless database used for its zero-configuration, ACID compliance, and suitability for research-scale deployment

- **Coverage Radius:** Each RSU monitors vehicles within a **500-meter radius** (typical for IEEE 802.11p).
- **Coverage and Redundancy:** The network design achieved **100%** vehicle coverage of the simulated routes. On average, each vehicle was detected by **3.2** RSUs, ensuring high data reliability through redundancy.
- **Data Collection:** Data is collected at a **10-second interval** during the simulation.

Table 3.6: RSU Detection Metrics (Partial Example)

RSU ID	Strategic Importance	Vehicles Detected	Records Collected (480s)
RSU_Doratana	Central hub, highest traffic	31	225
RSU_Palbari	Major residential area entry	30	185
RSU_Muroli	Peripheral terminal, low EV traffic	8	12

Data Collection and Analysis

The RSU software (simulated in SUMO) collects key telemetry: **position, speed, battery charge/capacity**, and **simulation timestamp**. The data is batched and transmitted to the server via **HTTP POST**. The `extract_data.py` script automates the retrieval of this data from the **SQLite database**, exporting it to **Excel and CSV files** 3.7 for external analysis. This process

includes calculating crucial statistics like **minimum, maximum, and average battery percentage**, which are essential inputs for the energy-efficient route optimization algorithms.

Table 3.7: Data Extraction from RSU (Partial Example)

RSU ID	Vehicle ID	Speed (m/s)	Battery Percentage	Collection Timestamp
RSU_Doratana	t_10_Easybike_ER-02B	7.1388065	49.464	2025-11-15T23:35:51Z
RSU_Doratana	t_10_Electric_Rickshaw_V8	1.6652936	49.9632	2025-11-15T23:35:51Z
RSU_Doratana	t_10_Small_Easybike_V12	4.4885627	49.6825	2025-11-15T23:35:51Z
RSU_Monihar	t_11_Easybike_ER-02B	7.5342046	49.41	2025-11-15T23:35:51Z
RSU_Monihar	t_11_Electric_Rickshaw_V8	2.0360149	49.9452	2025-11-15T23:35:51Z
RSU_Monihar	t_11_Small_Easybike_V12	5.3345936	49.5565	2025-11-15T23:35:51Z
RSU_Palbari	t_1_Easybike_ER-02B	7.6317725	49.397	2025-11-15T23:35:51Z
RSU_Palbari	t_1_Electric_Rickshaw_V8	1.6845218	49.9624	2025-11-15T23:35:51Z

Chapter 4

Results and Discussion

Chapter 4: Results and Discussion

This chapter presents the quantitative results obtained from the simulation of the Vehicle-to-Infrastructure (V2I) data collection system. It provides a detailed analysis of these results, discusses their significance in the context of the research objectives, and explores the broader implications for electric vehicle (EV) route optimization and infrastructure planning in Bangladesh. The presented findings reflect the data collected **till now**, validating the core architecture and providing the necessary dataset for subsequent analysis and algorithm development.

4.1 Results

The simulation was executed to model the behavior of **31 electric vehicles** over a **480-second** period within the defined road network of Jashore 4.1. The system, comprising **7 strategically placed Roadside Units (RSUs)**, a central FastAPI server, and a SQLite database, successfully collected real-time telemetry data.

4.1.1 Data Collection and System Performance

The primary objective was to validate the RSU-based data collection architecture. The system demonstrated robust performance, capturing a comprehensive dataset for analysis.

Table 4.1: Overall Data Collection Summary

Metric	Value
Simulation Duration	480.0 seconds
Total RSU Vehicle Records	959
Unique Vehicles Tracked	31 (100% of fleet)
Average Records per Second	2.0
Data-Generating RSUs	7

The `extract_data.py` script successfully processed the `telemetry.db` database, generating a timestamped Excel workbook (`rsu_telemetry_data_YYYYMMDD_HHMMSS.xlsx`) and corresponding CSV files, confirming the end-to-end data pipeline.

4.1.2 RSU Network Activity

The strategic placement of RSUs resulted in varied but comprehensive network coverage. RSU activity, measured by the number of records collected, highlights key traffic patterns and data collection hotspots 4.2.

Table 4.2: Data Records Collected per RSU

RSU ID	Location	Records Collected
RSU_Doratana	Doratana Intersection	225
RSU_Dhormotola	Dhormotola Road	206
RSU_Palbari	Palbari Junction	185
RSU_Chachra	Chachra Market	159
RSU_NewMarket	New Market	150
RSU_Monihar	Monihar Crossing	34
RSU_Muroli	Muroli Terminal	12
Total		959

As illustrated by the data in Table 4.2, RSU_Doratana and RSU_Dhormotola were the most active units, indicating their central position within the vehicle routes. In contrast, RSU_Muroli recorded the least activity, suggesting that the simulated routes rarely passed its location.

4.1.3 Battery Discharge and Energy Consumption

A critical outcome of this research was the successful modeling and measurement of realistic battery discharge 4.3. By configuring the vehicles with the collected Battery Device Parameters and using the Energy/unknown emission class in SUMO, the system captured dynamic energy consumption based on vehicle physics.

Table 4.3: Battery Percentage Statistics (Across All Vehicles)

Statistic	Battery Percentage (%)
Starting (Maximum)	50.00%
Minimum Recorded	37.80%
Average at End of Routes	45.70%

The results show a clear and measurable decrease in battery charge over the simulation period. A detailed analysis of a sample vehicle, t_10_Easybike_ER-02B 4.4, provides a granular

view of energy consumption.

Table 4.4: Sample Vehicle Discharge Analysis (t_10_Easybike_ER-02B)

Metric	Value
Vehicle Type	Easybike_ER-02B
Battery Capacity	3000 Wh
Starting Charge (50%)	1500.00 Wh
Final Charge (at 211s)	1224.10 Wh (40.80%)
Total Discharge (Wh)	275.90 Wh
Total Discharge (%)	9.20%
Route Duration	210 seconds

This represents a significant and realistic energy expenditure, providing the high-fidelity data necessary for energy-aware route optimization algorithms.

4.2 Discussion of Results

The results presented in Section 4.1 confirm the viability and effectiveness of the proposed V2I methodology. The discussion below interprets these findings in relation to the research goals.

4.2.1 Validation of the V2I Architecture

The successful collection of 959 records from 31 vehicles without data loss validates the end-to-end V2I architecture. The data flow from Vehicle → RSU → Server → Database performed as designed. The FastAPI server handled the concurrent data ingestion efficiently, and the batching mechanism was found to yield a **93%** reduction in per-record overhead. The system's ability to track **100%** of the active vehicle fleet demonstrates that the RSU network placement provides complete spatial coverage for the selected routes.

4.2.2 Significance of Battery Discharge Data

The most critical finding is the successful measurement of battery discharge, which is a direct result of configuring the vehicles accurately with battery device parameters in SUMO. The

collected dataset enables the calculation of crucial statistics like minimum, maximum, and average battery percentage, which are essential inputs for the energy-efficient route optimization algorithms. The 9.20% discharge observed for the Easybike over a short 210-second trip highlights a realistic energy consumption rate, which is a crucial factor for route planning where such vehicles are prevalent.

4.2.3 Analysis of RSU Network Effectiveness

The RSU activity data (Table 4.2) provides valuable insights into traffic flow and network design. The high activity at RSU_Doratana and RSU_Dhormotola confirms their status as major traffic arteries or "centroids" within the network. Conversely, the low activity at RSU_Muroli suggests it is either on a less-traveled route or at the periphery of the operational area.

This finding has two key implications:

1. **Network Redundancy:** On average, each vehicle was detected by **3.2** RSUs, ensuring high data reliability through redundancy.
2. **Infrastructure Planning:** For real-world deployment, this data suggests that RSUs at high-activity locations like Doratana are essential, informing cost-effective infrastructure planning.

4.3 Implications

The findings of this research have significant practical implications, providing a validated foundation for future work in smart mobility and EV adoption in Bangladesh.

4.3.1 Foundation for Energy-Efficient Route Optimization

The primary implication is that this system provides the foundational data required to build and test sophisticated route optimization algorithms. Traditional algorithms optimize for the

shortest or fastest path. However, the collected dataset, which correlates battery percentage with vehicle speed and location, enables the development of algorithms where the cost function is **energy consumption**.

4.3.2 Data-Driven Charging Infrastructure Planning

The RSU activity data serves as a direct proxy for EV traffic density. By analyzing hotspots like RSU_Doratana, urban planners and private companies can make **data-driven decisions** on where to install new charging stations. Placing chargers at these high-traffic locations maximizes their utilization and accessibility, mitigating "range anxiety" and encouraging further EV adoption.

4.3.3 Towards a Smart Traffic Management System for Jashore

The validated V2I system is a foundational component of a larger Intelligent Transportation System (ITS). The real-time vehicle location and speed data collected via TraCI and aggregated by the RSUs can be fed into a central traffic management center. This would allow authorities to monitor traffic flow and detect congestion in real-time, enabling potential interventions like dynamic signal adjustments.

4.3.4 Policy and Economic Implications

For a developing country like Bangladesh, this research proves that a robust V2I network can be built using open-source tools (SUMO, Python, FastAPI) and commodity hardware. This lowers the barrier to entry for government agencies and local startups to innovate in the smart mobility sector. The data can be used to formulate evidence-based policies, such as offering incentives for operating EVs or providing subsidies for charging stations in data-verified "charging deserts."

Chapter 5

Conclusion

Chapter 5: Conclusion

This chapter summarizes the key achievements of the research, outlines the limitations encountered during the implementation and simulation phase, and details the essential next steps required to complete the project, specifically focusing on the application of machine learning for energy-aware route optimization.

5.1 Concluding Remarks

This research successfully designed, implemented, and validated a robust Vehicle-to-Infrastructure (V2I) data collection architecture utilizing the Eclipse SUMO simulation environment. The core objectives were achieved through the creation of a custom road network of Jashore, Bangladesh, realistic configuration of three categories of battery-powered three-wheelers (Easybike ER-02B, Small Easybike V12, Electric Rickshaw V8), and the establishment of a functional data pipeline.

The key successes include:

- **V2I Architecture Validation:** The end-to-end system comprising SUMO simulation, TraCI data extraction, Roadside Units (RSUs), and a central FastAPI/SQLite server successfully collected 959 records from 31 vehicles over 480 seconds.
- **Complete Coverage and Redundancy:** The network design achieved **100%** vehicle tracking coverage over the simulated routes, with each vehicle detected by an average of **3.2** RSUs, ensuring high data reliability.
- **High-Fidelity Energy Data:** The simulation successfully measured dynamic battery discharge, with the fleet showing a measurable decrease in charge from an initial 50% to a minimum recorded of **37.80%**. This high-fidelity energy consumption data is the foundational output needed for energy-efficient routing.
- **Data-Driven Insights:** The RSU activity metrics clearly identified traffic centroids (RSU_Doratana and RSU_Dhormotola were the most active), providing direct insights for future infrastructure and charging station placement planning.

In conclusion, this phase of the research validated the methodology for acquiring realistic, time-series V2I telemetry data, laying a solid and verified foundation for the primary goal of developing energy-efficient route optimization algorithms.

5.2 Limitations

The current phase of the research, being pre-defense, is subject to the following limitations:

- **Simulation Scope:** The road network is limited to a custom creation of only **7 major junctions** of Jashore city, connected by 22 road edges. This is a simplification of the real urban environment.
- **Static Traffic Demand:** The traffic demand model relies on only **5 unique predefined vehicle routes** and configures all vehicles to depart simultaneously at 0.00s. This setup ensures comparative analysis but does not fully represent the dynamic, stochastic nature of real-world traffic flow.
- **Absence of Optimization/Prediction:** The project is currently limited to **data collection and validation**. The core outcome of the research—the implementation and evaluation of the energy-efficient route optimization algorithm has not yet been performed.
- **Simplified Charging Model:** While charging stations were placed strategically, the simulation did not model dynamic queueing or real-time charge priority mechanisms, only fixed power levels and a 5-second charge delay.

5.3 Next Steps To Do

To fully realize the potential of the collected dataset and complete the research objectives, the following steps are critical:

1. **Data Pre-processing and Feature Engineering:** The raw telemetry data (speed, battery, time) must be refined. This involves calculating derivative metrics such as instantaneous energy consumption rate (Watt-hours/km), rolling average speed, and segment-specific energy efficiency metrics, preparing the data for model ingestion.
2. **Machine Learning Model Selection and Training:** We will employ Machine Learning (ML) models to train on the extracted data. The primary goal of the ML models will be to accurately predict the energy consumption (Wh/km) for specific vehicle types across different road segments and under various driving conditions (speed, acceleration).
3. **Development of the Energy-Efficient Route Optimization Algorithm:** The energy prediction model will be integrated into a custom routing algorithm. This algorithm will use the predicted energy cost as the primary weight function (replacing simple distance or time) within a graph search framework. The goal is to identify the path that minimizes total energy expenditure while still meeting temporal constraints.
4. **Algorithm Evaluation and Comparison:** The developed energy-aware routing algorithm will be rigorously tested against the same simulated traffic scenarios. Its performance will be compared against traditional routing approaches using key metrics: total energy consumed, journey time, and final state-of-charge upon route completion.

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